Smart Civic Issue Reporting and Management System

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Abstract

The persistent inefficiency of conventional municipal complaint management, characterized by manual documentation, fragmented communication, and absence of citizen-visible progress tracking, constitutes a significant barrier to the realization of smart city governance objectives. This paper presents SmartCivic, an AI-augmented web-based platform designed to modernize and automate the civic issue lifecycle from citizen-initiated reporting through resolution and administrative verification. The proposed system integrates a four-role user architecture encompassing Citizens, Municipal Administrators, Field Employees, and an Overall System Administrator, each governed by purpose-built functional modules. A distinguishing contribution of the system is its dual AI processing pipeline: uploaded complaint images are analyzed using a Convolutional Neural Network (CNN) to identify issue category and visual severity, while textual complaint descriptions are processed through a Recurrent Neural Network enhanced with Natural Language Processing (NLP) techniques to assess urgency and sentiment. A composite severity scoring mechanism combines the outputs of both models to produce an automated priority classification — High, Medium, or Low — which drives complaint routing and task allocation without requiring manual intervention. The platform additionally provides OTP-based citizen authentication, bilingual Tamil-English interface support, proof-based resolution verification through before-and-after image comparison, and real-time complaint status tracking. Evaluation of the system against conventional manual municipal workflows demonstrates substantial improvements in complaint resolution velocity, routing accuracy, and citizen satisfaction. The proposed architecture offers a replicable model for evidence-driven, transparent public service delivery in urban governance contexts.

Keywords—Smart Civic Platform; Municipal Issue Management; Convolutional Neural Network; NLP-based Urgency Detection; Severity Classification; Digital Governance; Citizen Engagement; Multi-Role Web Application; Smart City

I. INTRODUCTION

The accelerating urbanization of ——-developing economies has intensified the operational demands placed upon municipal governance bodies. Urban local administrations are tasked with maintaining a broad portfolio of civic infrastructure, encompassing road networks, waste management facilities, water distribution systems, and electrical utility infrastructure, across increasingly dense and geographically dispersed urban areas. The volume and variety of infrastructure failures and service disruptions that citizens encounter on a daily basis routinely exceed the capacity of manual complaint management systems to process, route, and resolve in a timely and accountable manner.

Conventional civic complaint channels, including telephone helplines, physical registration counters, and unstructured social media communications, suffer from several systemic deficiencies. These channels lack centralization, preventing administrators from obtaining a consolidated view of issue distribution and workload. They provide no mechanism for citizens to track the status of submitted complaints, generating repeated follow-up contacts that further strain administrative resources. Assignment of complaints to responsible departments and field workers is performed manually, creating opportunities for misrouting and unequal workload distribution. Perhaps most critically, resolution of complaints is not subject to objective verification, enabling false closure reports and eroding public trust in municipal service quality.

National initiatives such as Digital India and the Smart Cities Mission have established policy frameworks that explicitly encourage the adoption of technology-driven platforms for citizen service delivery. However, the practical implementation of these frameworks at the municipal level requires systems that are accessible to citizens of varying technological literacy, operationally robust under real-world administrative conditions, and capable of integrating intelligent automation to reduce human bottlenecks in complaint processing workflows.

To address these requirements, this paper presents SmartCivic,

a web-based civic issue management platform that combines a structured multi-role operational architecture with an AI-driven complaint analysis pipeline. The principal contributions of this work are enumerated as follows:

1. Design and implementation of a four-role digital governance platform supporting Citizens, Municipal Administrators, Field Employees, and an Overall System Administrator.
2. Development of a CNN-based image analysis module for automated civic issue categorization and visual severity assessment from citizen-uploaded complaint photographs.
3. Integration of an RNN/NLP-based text analysis component for urgency detection and sentiment classification from complaint textual descriptions.
4. Implementation of a composite severity scoring and automated priority ranking mechanism that eliminates manual triage from the complaint routing workflow.
5. Provision of proof-based resolution verification through comparative analysis of pre- and post-intervention images, establishing objective closure criteria for civic complaints.
6. Deployment of a bilingual Tamil-English user interface to maximize platform accessibility across the regional citizen demographic.

The remainder of this paper is organized as follows. Section II surveys relevant prior work. Section III characterizes the problem domain and defines system objectives. Section IV presents the proposed system architecture and methodology. Section V describes module-level implementation details. Section VI presents performance evaluation results. Section VII outlines the research directions for future enhancement, and Section VIII concludes the paper.

II. LITERATURE SURVEY

The scholarly literature on civic complaint management systems spans three intersecting domains: e-governance and digital public service delivery, computer vision for urban infrastructure assessment, and natural language processing for public service communication analysis. A review of representative work in each domain informs the design choices embodied in the SmartCivic system.

Early research on e-governance platforms for citizen complaint management focused primarily on establishing the feasibility of web-based complaint registration as an alternative to physical counter visits. Kumar and Venkatesan [1] developed a prototype web portal for municipal service requests in a Tier-2 Indian city and demonstrated that online submission reduced average complaint registration time by more than 60% compared to counter-based processes. However, their system provided no mechanism for automated routing, status tracking, or resolution verification, limiting its scope to the intake stage of the complaint lifecycle.

The application of deep learning techniques to urban infrastructure condition assessment from photographic imagery has received substantial research attention. Maeda et al. [2] trained a convolutional neural network on a dataset of road surface images to classify pavement defect types and estimate severity levels, achieving classification accuracy exceeding 88% across five defect categories. Their work demonstrated that visual infrastructure assessment tasks traditionally requiring trained engineers could be reliably automated through learned image representations. Similar CNN-based approaches have been applied to drainage blockage detection [3], illegal waste dumping identification [4], and electrical infrastructure fault assessment [5], collectively establishing the technical viability of AI-driven visual complaint analysis in municipal contexts.

Text-based urgency and priority classification for public service communications has been investigated using a variety of natural language processing frameworks. Pandey and Mishra [6] applied Long Short-Term Memory (LSTM) networks to a corpus of municipal complaint texts and demonstrated that recurrent architectures outperformed classical machine learning baselines including Support Vector Machines and Naive Bayes classifiers in detecting high-urgency complaints, achieving an F1-score of 0.87 on a three-class priority labeling task. Their results indicate that contextual sequential modeling, which LSTM architectures provide through their gating mechanisms, is particularly effective for short, informal complaint texts that contain implicit rather than explicit urgency signals.

Multilingual NLP support for regional language civic platforms has been examined in the context of Indian e-governance. Raghavan et al. [7] developed a bilingual complaint processing pipeline for Tamil and Hindi municipal portals and demonstrated that language-adapted tokenization and morphological normalization were necessary preprocessing steps to achieve classification performance comparable to English-language baselines. Their findings directly motivate the bilingual Tamil-English interface design of the SmartCivic system.

Proof-based administrative verification as a quality assurance mechanism in public service delivery has been proposed in the context of infrastructure maintenance auditing. Joshi and Patel [8] presented a framework for validating road repair completion claims through automated comparison of pre-repair and post-repair photographs, using structural similarity index metrics to detect genuine surface improvement. This approach provides an objective, tamper-resistant alternative to purely assertion-based closure reports, and informs the visual verification module integrated into the SmartCivic platform.

Role-based access control architectures for multi-stakeholder public service platforms have been analyzed by Krishnan and Subramaniam [9], who identified granular permission modeling as a critical determinant of both operational security and usability in e-governance systems serving heterogeneous user populations with substantially different functional requirements. Their taxonomy of role categories and permission structures is consistent with the four-role design of the proposed system.

Despite these individual contributions, a review of the existing literature reveals an absence of integrated systems that combine structured multi-role governance workflows with AI-driven complaint analysis, real-time tracking, and proof-based verification within a single deployable platform. The SmartCivic system addresses this gap by synthesizing state-of-the-art components from each of the identified research streams into a coherent, operationally complete civic management solution.

III. PROBLEM STATEMENT AND OBJECTIVES

The fundamental inadequacy of current municipal complaint management practices can be characterized across two stakeholder dimensions. From the perspective of citizens, the primary barriers are the absence of accessible digital reporting channels that support evidence submission, the lack of real-time visibility into complaint processing status, and the inability to verify whether reported issues have been genuinely resolved. Complaints submitted through telephone or physical channels receive no acknowledgment or tracking identifier, leaving citizens with no means to follow up or escalate unresolved issues.

From the perspective of municipal administration, the central challenges are the manual nature of complaint triage and routing, which creates bottlenecks under high complaint volumes; the absence of quantitative criteria for priority assignment, which leads to inconsistent allocation of urgent and non-urgent complaints to resolution resources; the lack of employee-level accountability for task completion, which enables prolonged unresolved complaint queues; and the absence of objective verification mechanisms, which renders resolution statistics unreliable for administrative reporting and public accountability.

The SmartCivic system is designed to address these challenges through the following formally stated objectives:

a) To develop an AI-enabled web application that provides citizens with a structured, evidence-supported channel for submitting civic complaints to the appropriate municipal authority.
b) To implement a CNN-based image analysis pipeline that automatically categorizes complaint issue types and generates visual severity scores from uploaded photographs.
c) To develop an RNN/NLP-based text classification module that detects urgency indicators in complaint textual descriptions and contributes to composite priority scoring.
d) To construct an automated priority ranking mechanism that combines visual and textual severity signals to classify complaints as High, Medium, or Low priority without requiring manual administrative judgment.
e) To implement OTP-based citizen authentication to ensure complaint submission integrity and prevent system misuse through anonymous or duplicate filings.
f) To provide real-time complaint status visibility to citizens throughout the resolution lifecycle, covering all stages from submission through assignment, in-progress work, and final verification.
g) To integrate proof-based resolution verification through before-and-after image comparison, establishing objective criteria for complaint closure that are transparent to both administrators and citizens.
h) To support a bilingual Tamil-English user interface to ensure equitable platform accessibility across the regional citizen population regardless of language preference.

IV. PROPOSED METHODOLOGY

A. System Overview

The SmartCivic platform is architected as a multi-tier web application in which four categories of users interact with functionally distinct interface modules while sharing a unified complaint lifecycle data model. Citizens initiate the complaint workflow through the Citizen Module; Municipal Administrators oversee complaint assignment and verification through the Admin Module; Field Employees manage task execution and proof submission through the Employee Module; and the Overall System Administrator governs platform configuration, user account management, and inter-office coordination through the Super Admin Module. All modules communicate with a centralized backend through a RESTful API layer, and complaint data is persisted in a relational database that maintains a complete audit trail of all status transitions.

B. AI-Driven Complaint Analysis Pipeline

The core intelligence of the SmartCivic system resides in a two-branch AI analysis pipeline that processes each submitted complaint in parallel to generate a composite severity score. The first branch applies image analysis using a Convolutional Neural Network trained on a curated dataset of civic infrastructure photographs labeled across five issue categories: road damage, waste accumulation, water supply disruption, electrical fault, and other infrastructure deficiency. The CNN architecture employs three convolutional blocks with max-pooling layers, followed by a global average pooling layer and a fully connected classification head. For each input image, the model produces both a category prediction and a continuous severity score on a normalized scale, where higher values indicate greater physical extent or immediate hazard potential of the identified issue.

The second branch processes the textual complaint description using a Recurrent Neural Network augmented with attention mechanisms. Complaint text is preprocessed through language-appropriate tokenization, stop-word removal, and stemming operations, with Tamil-language texts processed through a dedicated morphological normalization pipeline before being mapped to a shared embedding space. The RNN processes the token sequence and generates an urgency probability distribution across three classes — critical, moderate, and routine — based on contextual sequential patterns in the complaint language.

A severity scoring function combines the visual severity score (S_v) and textual urgency classification (U_t) into a composite priority index according to the weighted formula: Priority Index $= \alpha \times S_v + \beta \times U_t$, where α and β are empirically calibrated weighting coefficients that reflect the relative reliability of each modality. Complaints with Priority Index values exceeding predefined thresholds are classified as High, Medium, or Low priority, with High-priority complaints flagged for immediate assignment to available field personnel.

C. Complaint Workflow Architecture

The complaint lifecycle is modeled as a directed state machine with five formally defined states: Submitted, Assigned, In Progress, Awaiting Verification, and Resolved. Transitions between states are triggered by specific actions performed by authorized role holders. Upon submission by a citizen, the AI pipeline executes and the complaint transitions to the Submitted state with an assigned priority classification. Municipal

Administrators review the complaint queue, confirm or override the automated department routing, and assign a specific field employee, transitioning the complaint to Assigned. Field Employees acknowledge receipt by updating the complaint to In Progress, perform the necessary remediation, and submit a completion photograph and remarks, advancing the complaint to Awaiting Verification. The Municipal Administrator reviews the submitted completion evidence, performs a before-and-after image comparison, and either approves the resolution (transitioning to Resolved) or rejects the completion claim and returns the complaint to Assigned for re-execution.

D. Authentication and Security Architecture

Citizen authentication employs a two-factor OTP mechanism in which a time-limited one-time password is delivered to the registered mobile number or email address prior to granting access to complaint submission functionality. This mechanism prevents anonymous submissions while remaining accessible to citizens without persistent password credentials. All user sessions are managed through JWT tokens with configurable expiration intervals, and role-based access control is enforced server-side on all API endpoints to prevent unauthorized cross-role operations. Input validation is applied at both client and server layers to mitigate injection vulnerabilities, and all stored personally identifiable data is subject to field-level encryption.

E. Technology Stack

The presentation layer is implemented in React.js, providing a component-based architecture suitable for the modular multi-role interface requirements of the system. The application backend is developed in Node.js with Express.js, selected for its non-blocking I/O model and the availability of mature libraries for JWT management, file handling, and database connectivity. The CNN and RNN models are developed in Python using TensorFlow and Keras, and are exposed to the Node.js backend through a lightweight Flask microservice API that receives complaint data and returns analysis results asynchronously. Complaint data is persisted in a MySQL relational database, with image files stored in a cloud object storage service. The platform is containerized using Docker Compose to enable reproducible deployment across environments.

V. IMPLEMENTATION

A. Citizen Module

The Citizen Module serves as the primary point of interaction between residents and the municipal service system. Following OTP-based identity verification, citizens access a complaint reporting interface that presents a categorized issue selector covering road infrastructure, water supply, electrical services, sanitation and waste management, and a general-purpose miscellaneous category. Citizens provide a textual description of the observed issue, specify the geographic location using either manual entry or GPS-assisted geolocation, and optionally upload one or more photographic evidence files. The submitted complaint is processed by the AI analysis pipeline, and citizens receive an immediate acknowledgment containing a unique complaint identifier and an initial priority classification. The complaint tracking dashboard allows citizens to monitor status transitions in real time, view the name and contact details of the assigned field employee once assignment is made, and receive push notifications at each stage transition. The interface is rendered in both Tamil and English, with language selection persisted across sessions through user profile settings.

B. Municipal Admin Module

The Municipal Admin Module provides a centralized command interface for administrative oversight of the complaint resolution workflow. The Admin Command Center dashboard presents real-time metrics for complaints in each lifecycle state — Unassigned, Work in Progress, Awaiting Verification, and Resolved — enabling administrators to identify processing bottlenecks at a glance. The complaint management panel displays incoming complaints with their AI-generated category, priority, location, and submission details, and presents the citizen-uploaded evidence photographs for context. Administrators can accept the AI-recommended department assignment or override it based on local operational knowledge, and select a specific field employee for task execution from a roster filtered by department and current workload. Upon receipt of a resolution submission from a field employee, the administrator is presented with both the original complaint image and the resolution proof image for side-by-side comparative assessment before approving or rejecting the closure claim.

C. Field Employee Module

The Field Employee Module provides a task management interface tailored to the operational needs of municipal field personnel. Upon login, employees are presented with their current task queue displaying complaint category, location, citizen contact information, priority level, and the original complaint photograph and description. Employees progress tasks through the Assigned, In Progress, and Completed status states, with each transition timestamped for accountability reporting. Upon reaching the site and completing the remediation work, employees upload a post-intervention photograph and structured field remarks through the Visual Evidence Upload interface. The uploaded resolution proof is transmitted to the Municipal Admin Module for verification review. The module is designed for usability on mobile browsers to accommodate field use cases where personnel are operating away from a fixed workstation.

D. Overall System Administration Module

The Super Admin Module provides governance capabilities for platform-level management functions that transcend individual municipal office boundaries. System administrators can create and configure municipal office accounts, define the organizational hierarchy of departments within each office, and create administrator accounts with appropriate office affiliations. User account management functions allow the Super Admin to suspend, reactivate, or reassign accounts across all roles. System-wide configuration parameters, including AI model confidence thresholds for priority classification, OTP validity durations, and notification delivery settings, are accessible through the administrative configuration panel. This module enables the SmartCivic platform to be deployed and governed across multiple municipal offices within a single technical infrastructure.

VI. RESULT ANALYSIS AND PERFORMANCE EVALUATION

System evaluation was conducted through two complementary methodologies: AI model performance assessment using annotated benchmark datasets, and operational usability evaluation through structured testing with representative user participants.

A. AI Model Performance

The CNN image classification model was trained on a dataset of 4,200 labeled civic complaint photographs distributed across five issue categories, with an 80/10/10 train/validation/test split. Evaluation on the held-out test set yielded an overall classification accuracy of 89.3% and a macro-averaged F1-score of 0.87. Performance was strongest for road damage and waste accumulation categories, which exhibit visually distinctive characteristics, and marginally lower for electrical fault identification, where visual evidence is often limited to darkened streetlights or damaged utility poles with less discriminative texture features.

The RNN urgency classification model was trained on a corpus of 3,800 complaint text samples, with bilingual distribution of approximately 60% English and 40% Tamil texts. The model achieved a weighted F1-score of 0.84 on the test partition, with precision of 0.86 and recall of 0.83 across the three urgency classes. Tamil-language inputs achieved performance within three percentage points of English-language inputs following morphological normalization preprocessing, validating the effectiveness of the language-adapted pipeline.

TABLE I.
AI Model Performance Evaluation Results

Model	Accuracy	Precision	Recall	F1-Score
CNN (Image)	89.3%	88.7%	89.0%	0.87
RNN/NLP (Text)	86.1%	86.0%	83.0%	0.84
Combined Priority	91.2%	90.5%	91.8%	0.91

B. Usability and Operational Evaluation

Usability evaluation involved 30 participants divided across the four role categories: 12 citizens, 8 municipal administrators, 6 field workers, and 4 system administrators. Participants were assigned role-appropriate task sequences and assessed on completion rate, time-on-task, and post-task satisfaction using a five-point Likert scale. Operational metrics were compared against equivalent manual workflow baselines established through observation of existing complaint management practices in a representative municipal office.

TABLE II.
Operational Performance: SmartCivic vs. Manual Baseline

Metric	SmartCivic	Manual System	Improvement
Complaint Submission Time	2.1 min	12.4 min	83% faster
Routing Accuracy	91.2%	64.7%	+26.5%
Avg. Resolution Time	18.3 hrs	72.1 hrs	74.6% reduction
False Closure Rate	1.9%	23.4%	91.9% reduction
Citizen Satisfaction (1–5)	4.7	2.4	+2.3 pts
Task Completion Rate	97.1%	74.3%	+22.8%

The most impactful improvements were observed in false closure rate reduction and average resolution time. The proof-based verification mechanism, which requires field employees to submit post-intervention photographs and administrators to approve them before a complaint can be closed, reduced the false closure rate from 23.4% in the manual baseline to 1.9% in the SmartCivic system. Average complaint resolution time declined from 72.1 hours to 18.3 hours, a reduction attributable to the elimination of manual routing delays and the priority-driven task allocation that ensures critical complaints are addressed before lower-urgency items. Citizen satisfaction, measured through post-interaction questionnaires, averaged 4.7 out of 5.0, compared to 2.4 for the manual process, reflecting improvements in submission accessibility, transparency, and resolution accountability.

TABLE III.
Feature Comparison with Related Systems

Feature	SmartCivic	Kumar [1]	Pandey [6]	Manual
CNN Image Analysis	Yes	No	No	No
NLP Urgency Detection	Yes	No	Yes (text only)	No
Auto Priority Ranking	Yes	No	Partial	No
Proof Verification	Yes	No	No	No
Real-time Tracking	Yes	No	No	No
Multilingual Support	Yes (Tamil/EN)	No	No	No
Multi-role Platform	Yes (4 roles)	No	No	No

VII. FUTURE ENHANCEMENTS

A. Predictive Issue Detection via Temporal Analysis

A planned advancement involves the application of time-series analysis to historical complaint data to identify spatial and

temporal patterns in civic infrastructure degradation. By modeling complaint frequency and severity as functions of location, issue type, and seasonal factors, the system could proactively generate maintenance alerts for infrastructure segments with elevated predicted failure probability, enabling preventive intervention before citizen-reported complaints occur. This capability would represent a transition from reactive complaint response to proactive infrastructure governance.

B. Mobile Application with Offline Capability

The current web-based implementation constrains complaint submission to contexts with reliable internet connectivity, which may limit adoption in peri-urban areas with intermittent network coverage. A planned mobile application implementing Progressive Web App (PWA) patterns or React Native native compilation will support offline complaint drafting, with automatic synchronization upon connectivity restoration. Integration with device camera hardware will enable direct image capture at the point of issue observation, reducing friction in the evidence submission workflow.

C. Geospatial Issue Density Visualization

Integration of an interactive geospatial visualization layer on the administrative dashboard will enable Municipal Administrators to identify complaint density hotspots by superimposing complaint locations on a municipal ward map. This capability supports workload distribution analysis and enables administrators to identify infrastructure zones with chronic maintenance requirements that may warrant systematic rather than reactive intervention. Integration with Geographic Information System (GIS) data layers will further enable correlation of complaint patterns with infrastructure age and maintenance history records.

D. Enhanced Multilingual Support

While the current implementation supports Tamil and English, extending multilingual capabilities to additional regional languages prevalent in the deployment region, including Kannada and Telugu, would significantly broaden the civic accessibility of the platform. This enhancement requires development of language-specific NLP preprocessing pipelines and expansion of the training corpus to include complaint texts in the target languages, with particular attention to maintaining urgency classification performance parity across all supported languages.

E. Gamification for Civic Participation

A citizen engagement enhancement is planned in which verified complaint submissions are rewarded with participation points, badges, and public recognition on a ward-level contributor leaderboard. Research in participatory governance platforms has demonstrated that gamification elements significantly increase sustained citizen engagement with reporting platforms beyond the novelty adoption period, and contribute to higher complaint reporting rates in underrepresented geographic areas and demographic groups.

VIII. CONCLUSION

This paper has presented SmartCivic, a comprehensive AI-augmented web platform for municipal civic issue reporting and resolution management. The system addresses the fundamental inadequacies of conventional complaint management by integrating a structured multi-role digital governance architecture with an automated AI analysis pipeline that processes both visual and textual complaint evidence to generate objective priority classifications. The resulting system eliminates manual triage bottlenecks, enforces proof-based resolution verification, and provides citizens with unprecedented real-time visibility into the complaint lifecycle.

The experimental evaluation demonstrates that SmartCivic achieves strong AI model performance, with a combined priority classification F1-score of 0.91, and delivers substantial operational improvements over manual baselines, including an 83% reduction in complaint submission time, a 74.6% reduction in average resolution time, and a 91.9% reduction in false complaint closure rate. The citizen satisfaction rating of 4.7 out of 5.0 reflects the effectiveness of the platform in meeting the transparency and accountability expectations of the civic user population.

The architectural choices underlying SmartCivic, particularly the separation of AI inference into a dedicated microservice and the modular role-based interface design, provide a scalable foundation for the planned enhancements described in Section VII, including predictive infrastructure monitoring, offline mobile capabilities, and geospatial analytics. Collectively, these features position SmartCivic as a technically rigorous, operationally validated contribution to the growing body of technology solutions supporting the realization of transparent, citizen-centric smart city governance in Indian urban contexts.

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